

**Task ‘Object-Oriented Software Architecture and Test-Driven Development for
an E-Learning Platform’ Unit 10**

Secure E-Learning Platform: Design, Implementation, and Testing

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Introduction

This report demonstrates the design, implementation, and testing of a secure prototype-level e-learning platform that uses object-oriented principles and Test-Driven Development (TDD), the platform addresses maintainability, defensive programming, and input validation while demonstrating established software engineering practices in architecture and testing methodologies. TDD supports code quality by supplying protection for refactoring and promoting a practice to identify defects earlier in the development lifecycle. The report also critically evaluates architectural and testing decisions, recognizing limitations associated with prototype-scale systems.

(Beck, 2021)

Task 1: Software Architecture Design

Architectural Style Selection

The selection of Monolithic layered architecture over microservices architecture was done due to lower operational complexity and practicability for a prototype-scale educational platform. Monolithic systems simplify deployment, development, and transaction management due to the fact that components generally operate within the same application boundary.

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Such approach is more appropriate for small-to-medium systems where rapid development, minimal cost and maintainability are prioritized over distributed scalability. While monolithic architecture helps reduce infrastructure complexity, some trade-offs are involved, and it may become more difficult to scale and maintain as system size increases because of the tighter coupling dependency of components.

(Simion, 2023)

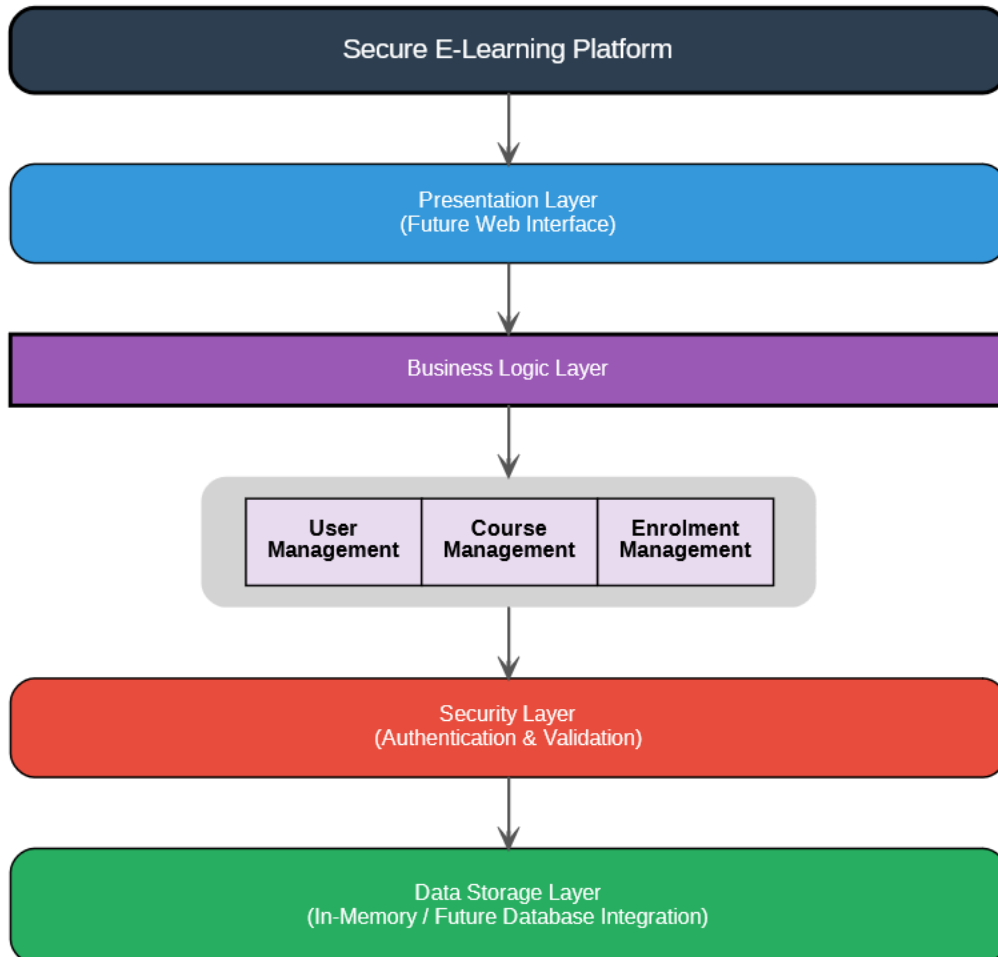
On the other hand, microservices support independent scaling deployment, these may also introduce additional orchestration and communication complexity, therefore the selected architecture should be viewed contextually appropriate for a use case, rather than universally. Input validation and business-rule enforcement are both centralized within the application layer, which supports consistency in validation logic and reduces duplicated checks across modules.

(Khan, et al. 2021)

High-Level Architecture Diagram

Figure 1: High-Level Architecture Diagram

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Core Modules

The system designed into three primary modules (User Management, Course Management and Enrolment Management), with such separation representing a logical decomposition of domain responsibilities that supports the Single Responsibility Principle and improves maintainability, the User Management module provides a fundamental framework for future authentication and authorization functionality, while Course Management handles course creation, retrieval, instructor allocation, and Enrolment Management oversees student registration processes and

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enforces business rules related to capacity and duplication restrictions. This modular separation is appropriate at a conceptual level, the current design remains tightly coupled within a monolithic boundary, as system complexity increases, this may lead to increased interdependency between modules, reducing maintainability unless further abstraction layers are introduced.

Security Considerations

The Security implementation follows the secure software engineering principles. Password storage will utilize adaptive hashing algorithms such as bcrypt, PBKDF2, or Argon2, where Argon2 is regarded as the best modern technique due to the GPU-accelerated attacks resistance, authentication design proposes role-based access control (RBAC) to ensure permissions separation between students, instructors, and administrators and input validation applies defensive programming practices by validating all user inputs against expected formats and ranges. Parameterized queries or secure ORM frameworks would be required to mitigate injection vulnerabilities Where database integration is introduced. Secure-by-design approaches would require validation and governance mechanisms to be integrated throughout the software lifecycle rather than treated as isolated security features. (OWASP Foundation 2026).

Task 2: Implementation

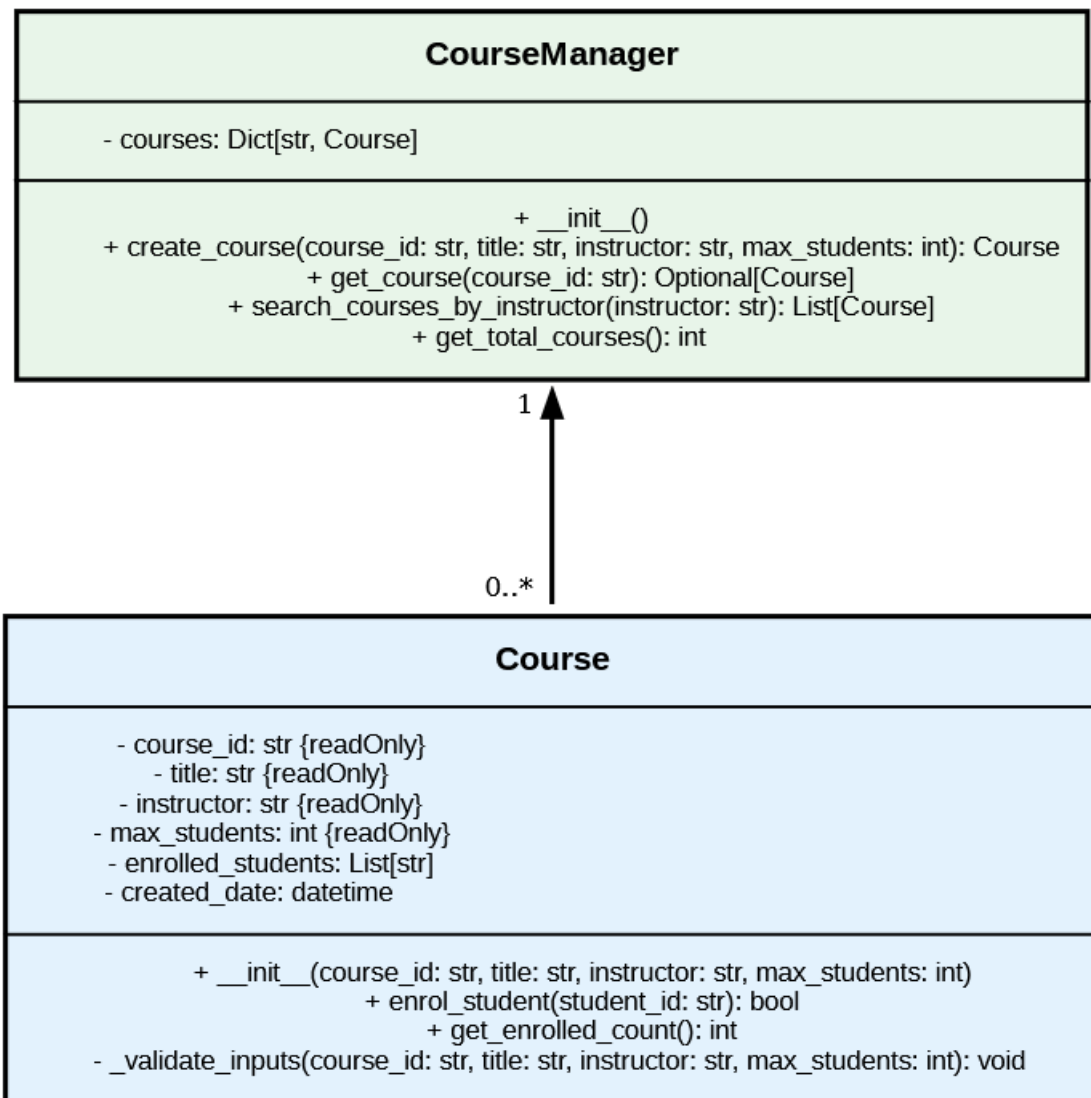
Object-Oriented Implementation

The Course Management module demonstrates core OOP principles including encapsulation and data validation. The implementation includes Python type hints

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and property decorators to control attribute access whilst maintaining clear interfaces. To represent the object-oriented structure of the implemented Course Management module, a UML class diagram was developed showing the primary classes, attributes, methods, and aggregation relationships used within the business logic layer.

Figure 2: UML Class Diagram for the Course Management Module



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The UML class diagrams have been created using Mermaid and PlantUML.

ChatGPT (OpenAI, 2026) was used to assist in generating and refining the Mermaid and PlantUML code prior to rendering the diagrams

(Mermaid, 2026; OpenAI, 2026; PlantUML, 2026)

Design Interpretation of UML Model

The UML class diagram illustrates the structural design of the Course Management module within the business logic layer, the system consists of two primary classes, Course and CourseManager, which implement enrolment and course administration functionality. The Course class encapsulates course-related data and validation logic maintaining the state consistency through controlled attribute access, Read-only properties support immutability for key identifiers where modification after instantiation would result inappropriate and the CourseManager class corresponds multiple Course instances and provides creation, retrieval, and search operations using an in-memory dictionary structure, such aggregation relationship reflects a “has-a” association where CourseManager is maintaining the references to Course objects without directly controlling their lifecycle, additionally the design supports separation of concerns and aligns with encapsulation and the single-responsibility principles.

(Simion, 2023).

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Representative Unit Test Examples

Code Snippet 1: Unit Test for Invalid Course ID Validation

```
```python
def test_invalid_course_id_raises_error(self):
 with self.assertRaises(ValidationError):
 Course("invalid", "Title", "Instructor", 30)
...```
```

#### Code Snippet 2: Unit Test for Duplicate Enrolment Prevention

```
```python
def test_duplicate_enrolment_prevented(self):
    course = Course("CS101", "Python", "Dr Smith", 30)
    course.enrol_student("STU12345")
    result = course.enrol_student("STU12345")
    self.assertFalse(result)
...```
```

Code Snippet 3: Student ID Validation in enrol_student()

```
```python
class Course:
 def enrol_student(self, student_id: str) -> bool:
...```
```

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```
if not re.match(r'^STU[0-9]{5}$', student_id):
```

```
 raise ValidationError("Invalid student ID format")
```

```
...
```

### Code Snippet 4: Case-Insensitive Instructor Search

```
```python
```

```
def search_courses_by_instructor(self, instructor: str) -> List[Course]:
```

```
    return [c for c in self._courses.values()
```

```
        if c.instructor.lower() == instructor.lower()]
```

```
...
```

After writing the tests, the implementation was developed to satisfy each test case, embodying the red-green-refactor cycle central to TDD

(Simion, 2023).

Task 3: Testing and Validation

Test-Driven Development (TDD) Overview

Following the principles of Test-Driven Development (TDD), unit tests were written before implementation to specify the expected system behavior and guide development, the complete test suite is provided separately in the file `test\_course\_management.py`, representative examples are displayed below.

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Testing Strategy

The testing strategy focuses predominantly on unit testing, where Python's unittest framework from was used to run unit tests and validate individual components in isolation, confirming correctness of core business logic. The Course and CourseManager classes and TDD principles tests, were implemented before the active code development in order to define expected system behavior and guide implementation decisions, while unit testing provides strong component-level assurance, integration testing between modules has not been implemented in this prototype-level design, therefore, interactions between components are assumed, rather than fully validated under realistic workflows.

(Python Software Foundation, 2026).

As a future enhancement, integration testing and automated coverage analysis could improve confidence in system-wide correctness and testing completeness.

Test Execution Process

The unittest framework was executed in verbose mode to provide detailed test results. The command used was:

```
```bash (My Use Case)
```

```
PS C:\Users\AmnonMalka> cd .\Documents\
```

```
PS C:\Users\AmnonMalka\Documents> cd .\UOEO\
```

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```
PS C:\Users\AmnonMalka\Documents\UOEO> &
"C:\Users\AmnonMalka\AppData\Local\Programs\Python\Python314\python.exe" -m
unittest test_course_management.py -v
...

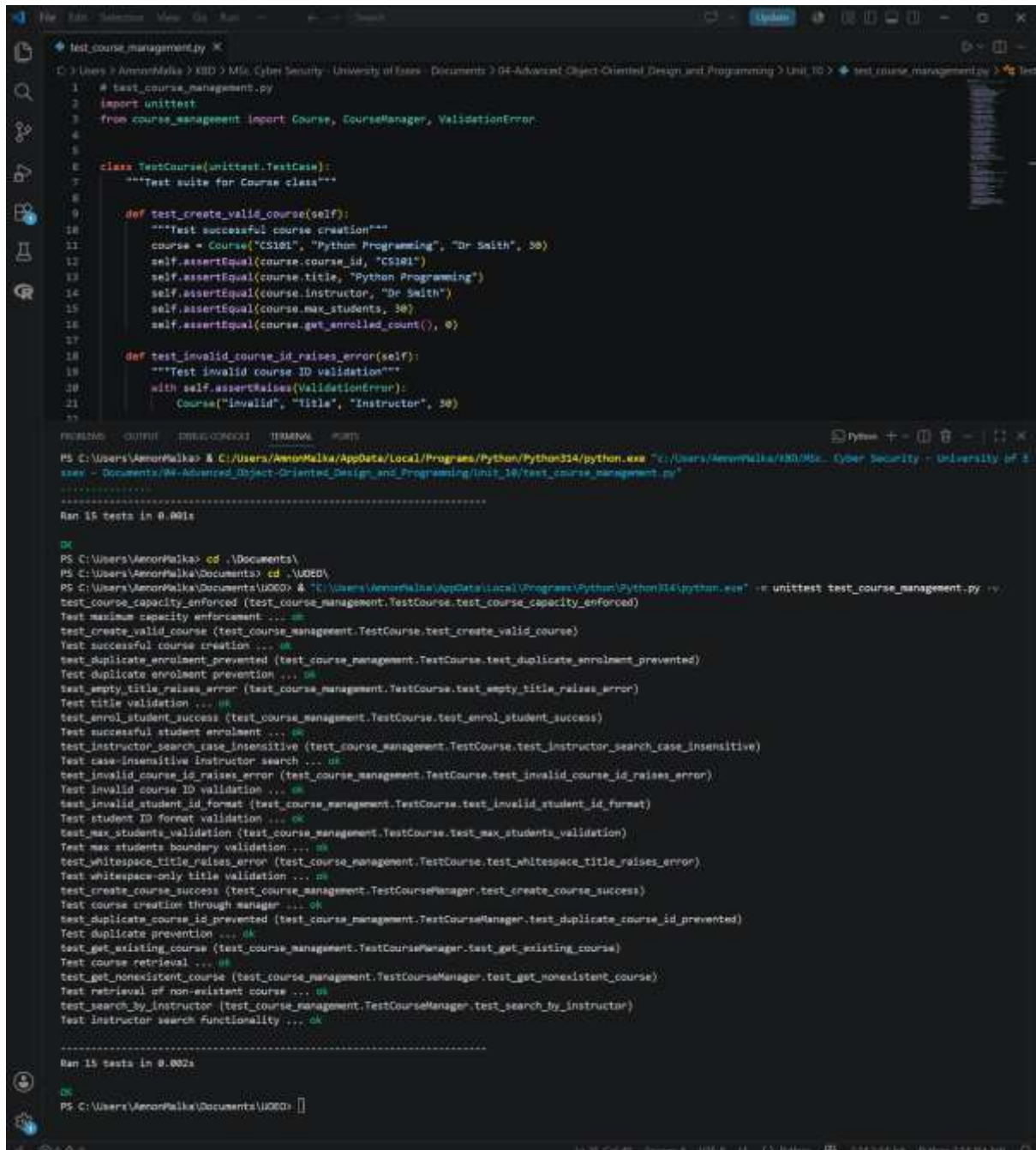
```bash (Simplified Use Case)

& "C:\Users\"Your User"\AppData\Local\Programs\Python\Python314\python.exe" -
m unittest test_course_management.py -v
...
```

Test Results

Figure 3: Terminal output showing all tests passing

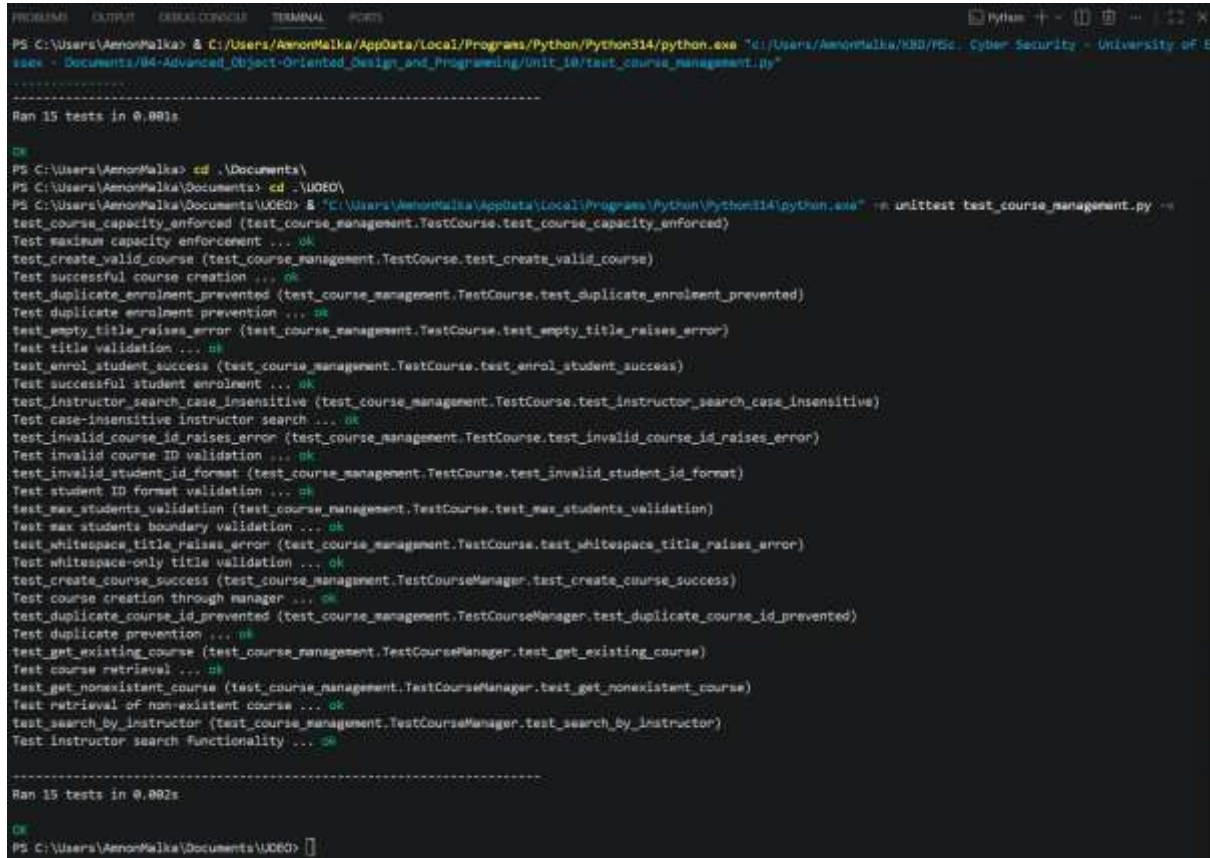
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```
test_course_management.py
1 # test_course_management.py
2 import unittest
3 from course_management import Course, CourseManager, ValidationError
4
5
6 class TestCourse(unittest.TestCase):
7     """Test suite for Course class"""
8
9     def test_create_valid_course(self):
10         """Test successful course creation"""
11         course = Course("CS101", "Python Programming", "Dr Smith", 30)
12         self.assertEqual(course.course_id, "CS101")
13         self.assertEqual(course.title, "Python Programming")
14         self.assertEqual(course.instructor, "Dr Smith")
15         self.assertEqual(course.max_students, 30)
16         self.assertEqual(course.get_enrolled_count(), 0)
17
18     def test_invalid_course_id_raises_error(self):
19         """Test invalid course ID validation"""
20         with self.assertRaises(ValidationError):
21             Course("invalid", "title", "Instructor", 30)
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Figure 4: Unit Test Execution Results



```
PS C:\Users\AmonMalka> & C:\Users\AmonMalka\AppData\Local\Programs\Python\Python314\python.exe "c:\Users\AmonMalka\KBD\PSD - Cyber Security - University of B
test - Documents\04-Advanced_Object-Oriented_Design_and_Programming\Unit_10\test_course_management.py"
-----
Ran 15 tests in 0.001s

OK
PS C:\Users\AmonMalka> cd .\Documents\
PS C:\Users\AmonMalka\Documents> cd -\UOED\
PS C:\Users\AmonMalka\Documents\UOED> & "C:\Users\AmonMalka\AppData\Local\Programs\Python\Python314\python.exe" -u unittest test_course_management.py -v
test_course_capacity_enforced (test_course_management.TestCourse.test_course_capacity_enforced)
Test maximum capacity enforcement ... ok
test_create_valid_course (test_course_management.TestCourse.test_create_valid_course)
Test successful course creation ... ok
test_duplicate_enrolment_prevented (test_course_management.TestCourse.test_duplicate_enrolment_prevented)
Test duplicate enrolment prevention ... ok
test_empty_title_raises_error (test_course_management.TestCourse.test_empty_title_raises_error)
Test title validation ... ok
test_enrol_student_success (test_course_management.TestCourse.test_enrol_student_success)
Test successful student enrolment ... ok
test_instructor_search_case_insensitive (test_course_management.TestCourse.test_instructor_search_case_insensitive)
Test case-insensitive instructor search ... ok
test_invalid_course_id_raises_error (test_course_management.TestCourse.test_invalid_course_id_raises_error)
Test invalid course ID validation ... ok
test_invalid_student_id_format (test_course_management.TestCourse.test_invalid_student_id_format)
Test student ID format validation ... ok
test_max_students_validation (test_course_management.TestCourse.test_max_students_validation)
Test max students boundary validation ... ok
test_whitespace_title_raises_error (test_course_management.TestCourse.test_whitespace_title_raises_error)
Test whitespace-only title validation ... ok
test_create_course_success (test_course_management.TestCourseManager.test_create_course_success)
Test course creation through manager ... ok
test_duplicate_course_id_prevented (test_course_management.TestCourseManager.test_duplicate_course_id_prevented)
Test duplicate prevention ... ok
test_get_existing_course (test_course_management.TestCourseManager.test_get_existing_course)
Test course retrieval ... ok
test_get_nonexistent_course (test_course_management.TestCourseManager.test_get_nonexistent_course)
Test retrieval of non-existent course ... ok
test_search_by_instructor (test_course_management.TestCourseManager.test_search_by_instructor)
Test instructor search functionality ... ok

-----
Ran 15 tests in 0.002s

OK
PS C:\Users\AmonMalka\Documents\UOED>
```

Test Coverage Analysis

The test suite uses multiple testing techniques to verify system behavior and provide adequate coverage assurance of functional requirements.

- **Boundary Value Testing**

Validates capacity limits at boundaries (0, 1, 500, 501 students), ensuring the system correctly handles edge cases in accordance with specification requirements.

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- **Equivalence Partitioning**

Tests both valid inputs (properly formatted course IDs like "CS101") and invalid inputs (malformed IDs like "invalid") to verify input validation logic handles different input classes appropriately.

- **Negative Testing**

Ensures the proper handling of errors through `ValidationError` exceptions for scenarios including empty titles, whitespace-only titles, invalid student ID formats, duplicate enrolments, and exceeding course capacity.

State Verification

Confirms enrolment accuracy tracking by verifying student counts increment correctly and duplicate prevention mechanisms function as intended. Formal code coverage metrics were not collected as part of this prototype implementation, therefore testing completeness is evaluated qualitatively through representative unit, boundary and negative test cases rather than quantitatively through coverage analysis.

Input Validation

This approach enforces stricter input validation rules that was designed to improve data integrity and consistency within the secure architecture system.

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- **Format Validation**

Course IDs must match the regular expression pattern `^[A-Z]{2,4}[0-9]{3}$`, for instance CS101, MATH203, ensuring consistent identifier formats. Student IDs follow pattern `^STU[0-9]{5}$`, maintaining format consistency.

- **Boundary Enforcement**

Maximum student limits (1-500) prevent unrealistic enrolment values. Course capacity enforcement prevents over-enrolment in the business logic layer.

- **Duplicate Prevention**

Tests verify the system rejects duplicate course IDs and duplicate student enrolments, maintaining data integrity.

Refactoring and Improvements

While initial implementations were refactored following test feedback in line with the TDD "red-green-refactor" cycle, the validation logic was also extracted into reusable methods to improve maintainability and reduce duplication, as well as to promote code hygiene.

(Simion, 2023).

Private attributes and property decorators supported encapsulation while keeping interfaces under control and, additionally, the custom `ValidationError` exception class improved error categorization and debugging efficiency through clarity with such refactoring activities maintaining all passed tests and demonstrating TDD support for

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an improved and maintainable software evolution through continuous regression validation.

(Beck, 2021)

Limitations

For demonstration purposes, this prototype design uses in-memory storage, however, in a production environment, persistent databases such as SQLite or PostgreSQL, as well as transaction management and concurrency control mechanisms should be used. Additional enhancements should include authentication, role-based authorization, integration testing, and secure database interaction through parameterized queries. Although the current implementation presents core object-oriented and TDD principles effectively, the prototype remains limited in scalability in some enterprise-level design consideration, due to the absence of persistence and distributed deployment considerations.

Conclusion

This implementation helps demonstrate a maintainable e-learning prototype level platform developed using TDD and object-oriented software design principles. The design employed a monolithic layered architecture that provides a suitable structure for an education environment. Additionally, simplifying the implementation of business logic through the use of centralized validation mechanisms, input

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validation, format verification, and capacity enforcement, which help maintaining the logical consistency within the business logic layer. The TDD methodology created robust, well-documented code that supports improved reliability and maintainability. 15 unit tests covering multiple validation and enrolment scenarios validate core functionality and serve as executable documentation.

(Simion, 2023).

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Appendix A: Submitted Project Files

1. `course_management.py` – Core implementation
2. `test_course_management.py` – Unit test suite
3. `course_management_with_demo.py` – Demonstration script

Appendix B: Supporting Documentation

1. `README.md` – Instructions for executing the project and unit tests